NETWORK TRAFFIC FORENSICS

Malware Delivery & Windows Incident Investigation with Wireshark

Technical case study | Author: Pedro Gonçalves

PCAP analysis • DHCP / NBNS / Kerberos / HTTP • Malware delivery • Indicators of compromise • Evidence assessment

Executive summary

This investigation examines a Windows compromise recorded in a packet capture (PCAP). The analysis uses Wireshark to identify the affected endpoint, the active account, the suspicious delivery infrastructure and subsequent HTTP communications. Original laboratory screenshots are reproduced as supporting evidence, with the analytical conclusions and limitations stated separately.

Scope: retrospective examination of captured network traffic. The original PCAP and extracted executable were not supplied with this report; therefore hashes and attribution claims are transcribed from the source report and cannot be independently recalculated here.

1. Objectives and evidence handling

The case study addresses seven questions: endpoint IP/MAC/hostname; Windows user account; suspected delivery IP; executable extraction and SHA-256; malware classification; delivery host; and post-compromise domain. The workflow combines packet filters, protocol inspection, TCP-stream reconstruction and HTTP object export.

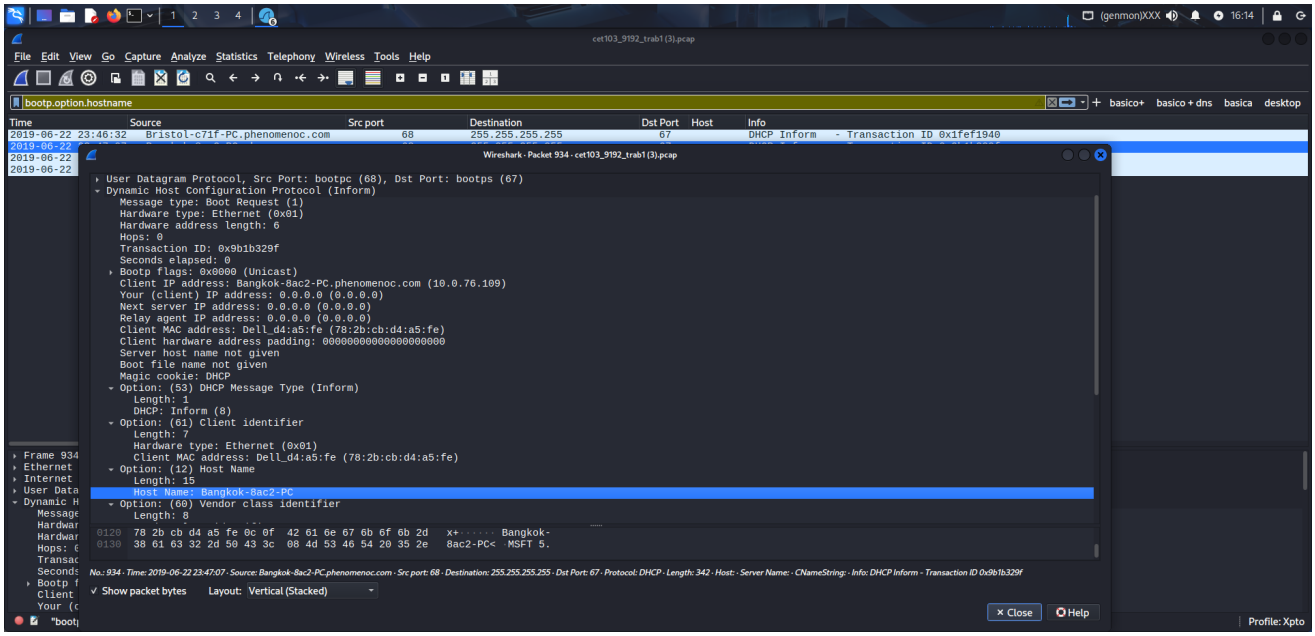
The original source includes differing SHA-256 values. These are explicitly retained as conflicting observations, not merged into a single verified hash. No live execution of the suspected malware is needed to inspect the capture.

2. Investigation workflow

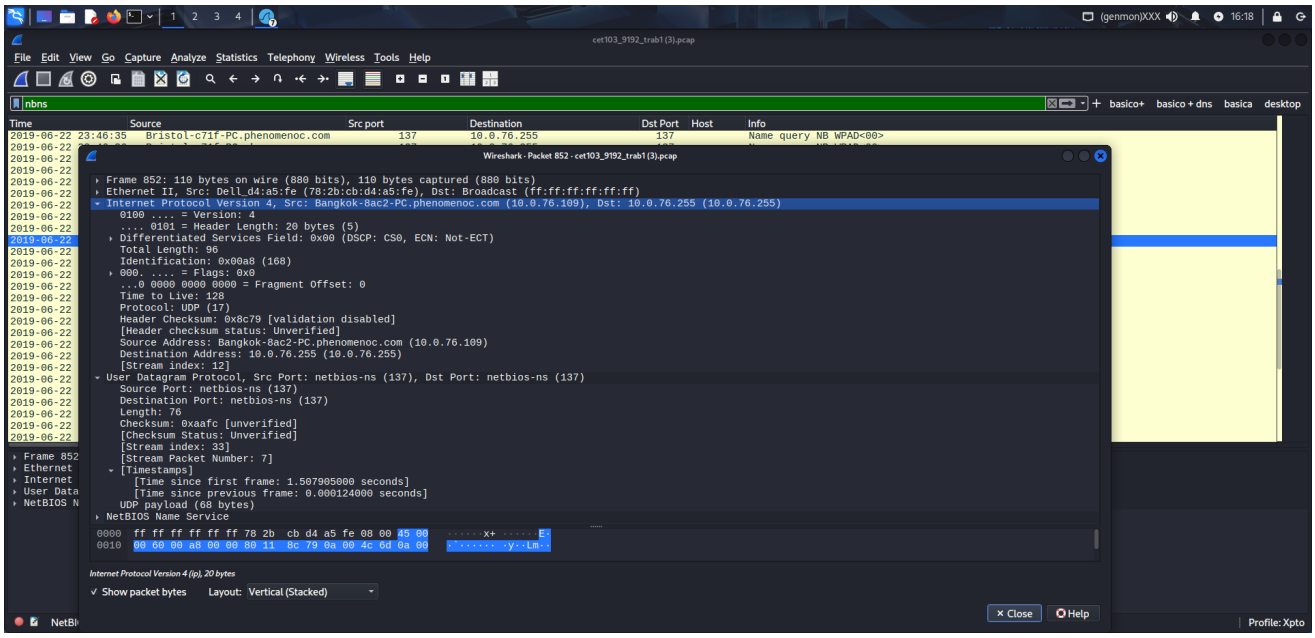
2.1 Identify the compromised endpoint using DHCP, NBNS and Ethernet headers

The DHCP hostname option provides a device name; NBNS/LLMNR/SMB traffic can corroborate that identity. Inspecting the IPv4 source and Ethernet II source fields links the network address to the device MAC address. The source report records hostname **BANGKOK-8AC2-PC**, IPv4 **10.0.76.109** and MAC **78:2b:cb:d4:a5:fe**.

Useful Wireshark display filters: `bootp.option.hostname`, `nbns`, `llmnr`, and `smb2`. A DHCP hostname alone should not be treated as proof of compromise; it establishes endpoint identity within the captured environment.



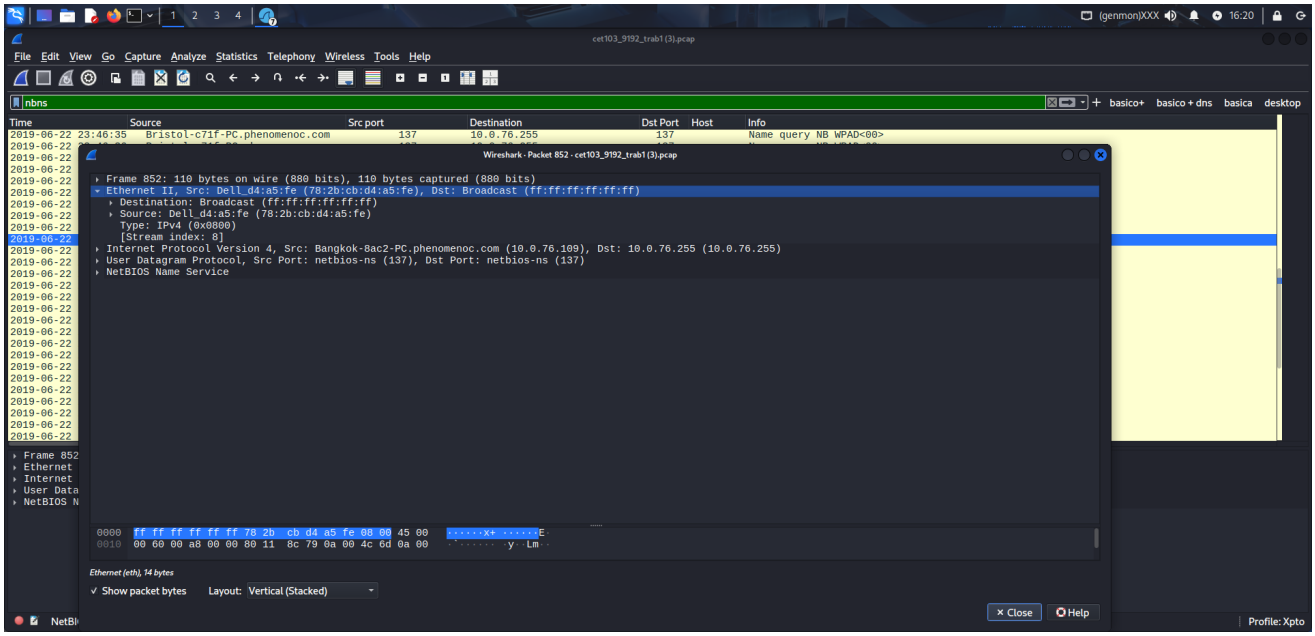
Evidence 01 — DHCP packet and hostname inspection



Evidence 02 — Packet detail inspection used to corroborate IPv4 and MAC fields

2.2 Identify the Windows user account

The report examines Kerberos authentication exchanges, including the client-name field, to associate observed network activity with a Windows account. The account reported is **edris.haight**. This is a network-derived account identifier and does not independently prove who was physically operating the workstation.

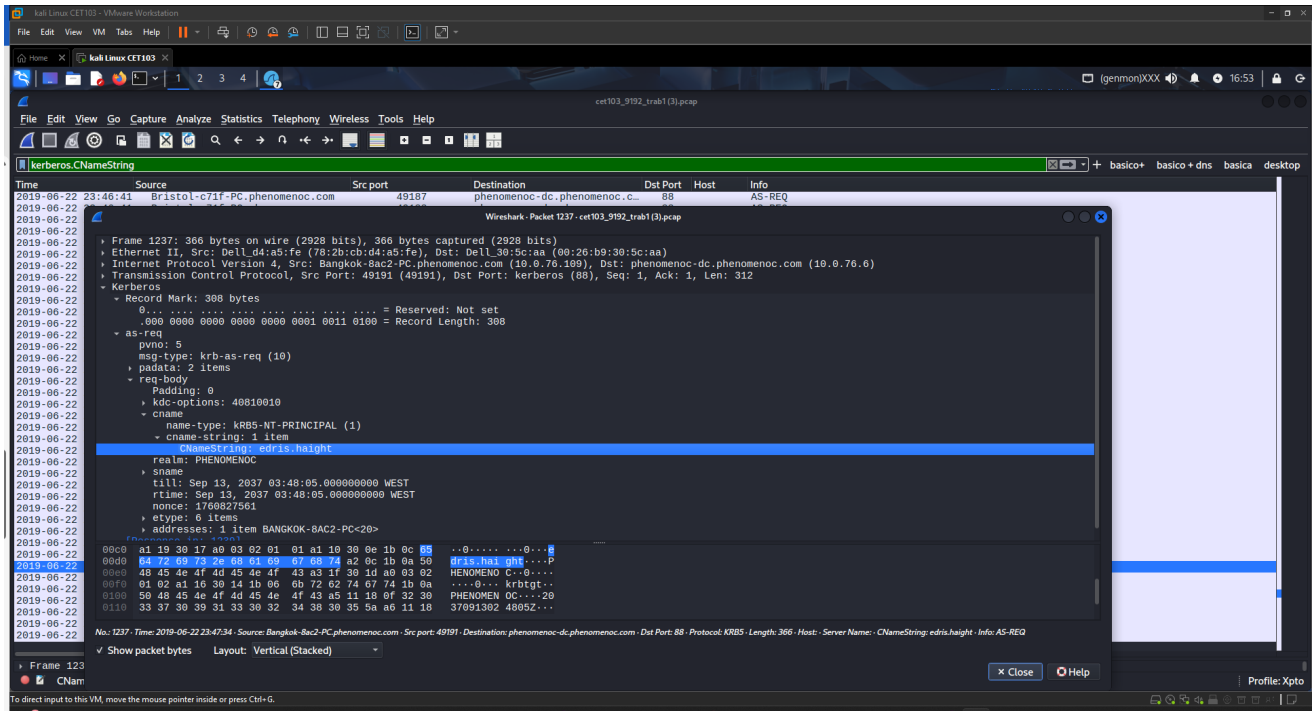


Evidence 03 — Kerberos and client identity inspection

2.3 Trace suspicious HTTP delivery

The analyst filtered HTTP requests and inspected suspicious GET requests with long query strings. A Follow TCP Stream view can show request/response context and an HTTP 200 OK response, supporting identification of the download source. The reported malware-delivery IP is **37.46.135.170**.

The source characterizes this activity as associated with Rig Exploit Kit. This is a reported case-study classification; independent verification would require the full PCAP and, ideally, IDS signatures or corroborating threat intelligence.

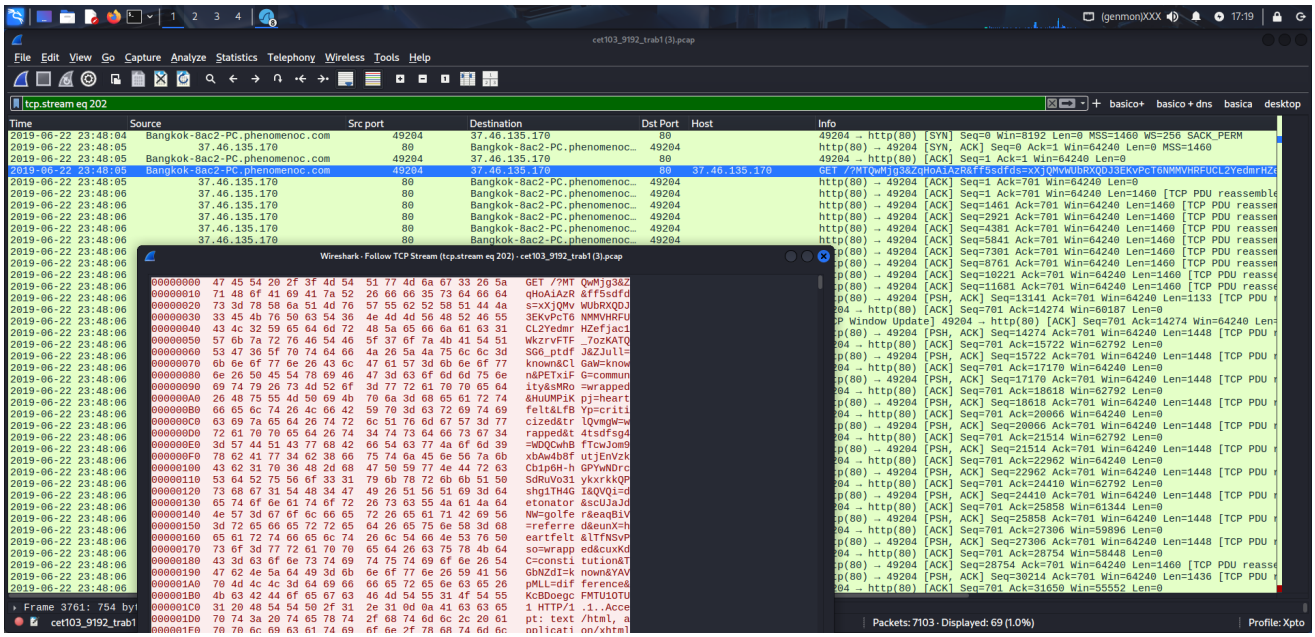


Evidence 04 — HTTP traffic inspection and suspected delivery stream

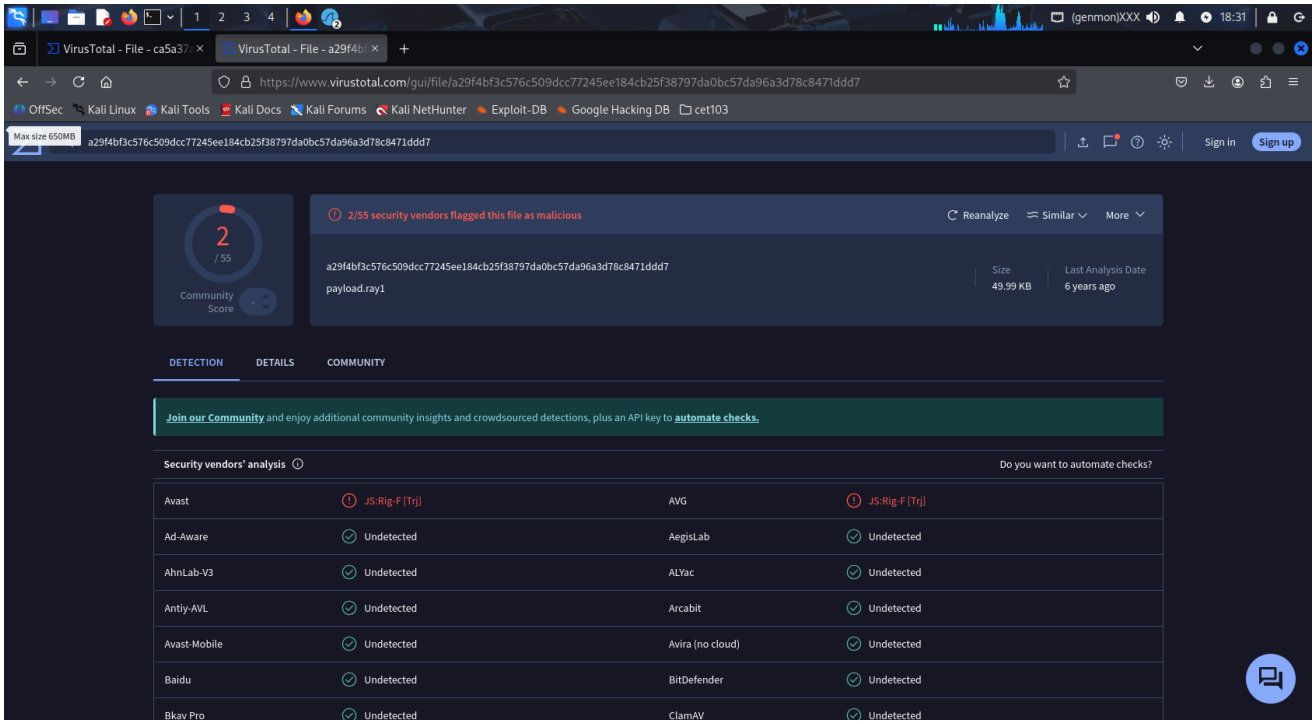
2.4 Examine downloaded content and its hashes

Wireshark can export HTTP-delivered objects through **File** → **Export Objects** → **HTTP**. The exported file should be retained as evidence, not executed, and hashed with a trusted SHA-256 utility. A hash links a specific byte sequence to an investigation; it does not by itself establish malware family or maliciousness.

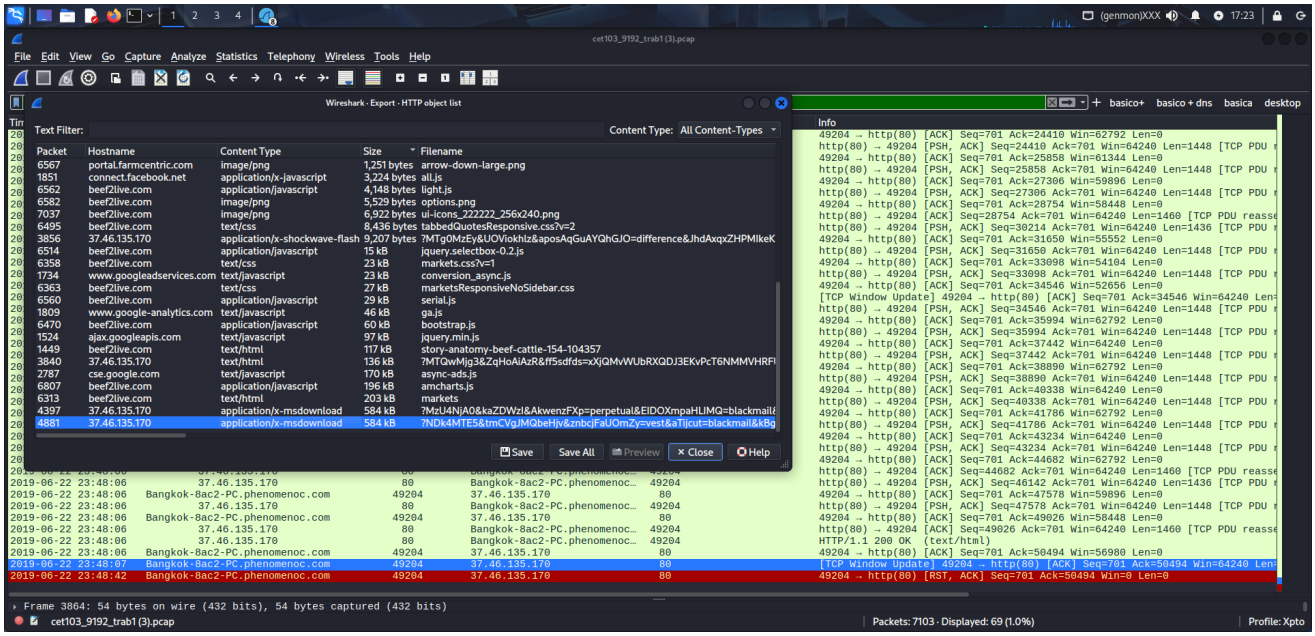
Two different hashes are shown in the source document: `a29f4bf3c576c509dcc77245ee184cb25f38797da0bc57da96a3d78c8471ddd7` and `ca5a37a5c3401ffcd1b7c98c3a22a921c013d1121fe33122e94dd81c382bf9b0`. A third hash is cited in the source summary: `39be5610259ffade85599720ee0af31187788a00791f1e4cb0cd05ef00105eda`. These are **not interchangeable**; the precise relationship between them cannot be resolved from the document alone.



Evidence 05 — Packet stream and candidate file hash



Evidence 06 — HTTP object export and evidence selection

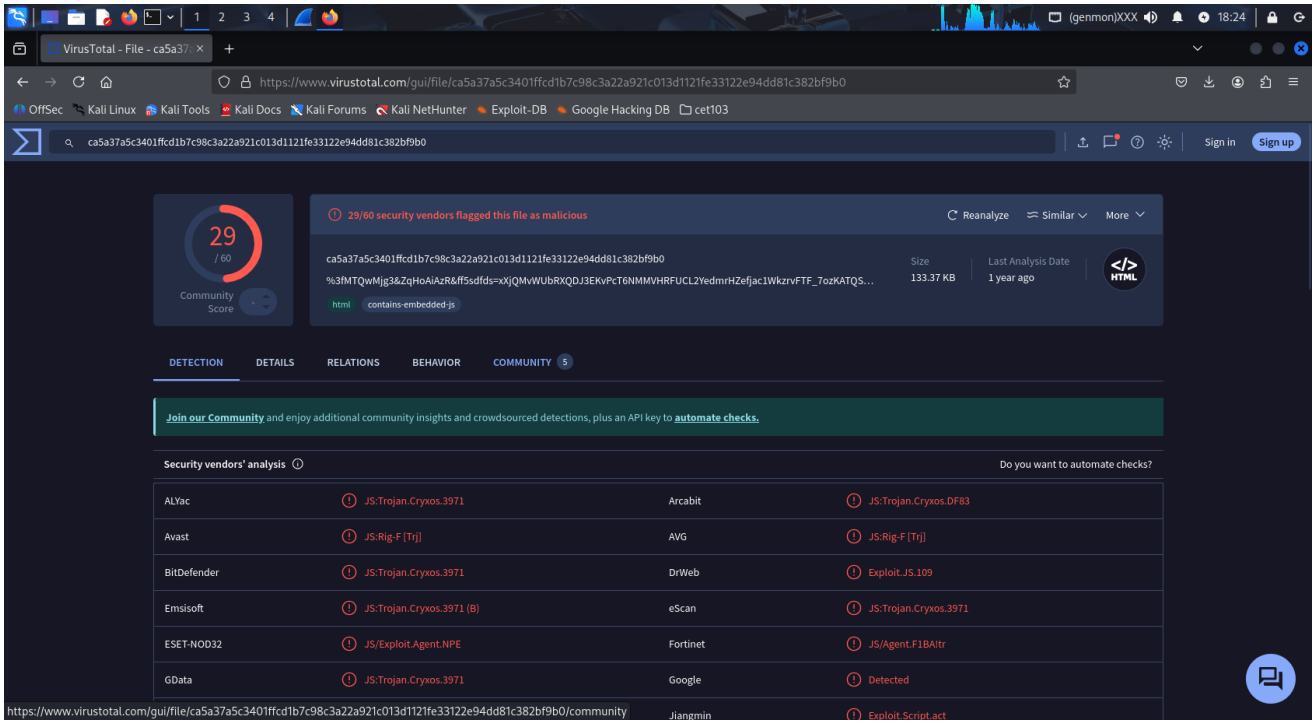


Evidence 07 — Threat-intelligence screenshot used in the original report

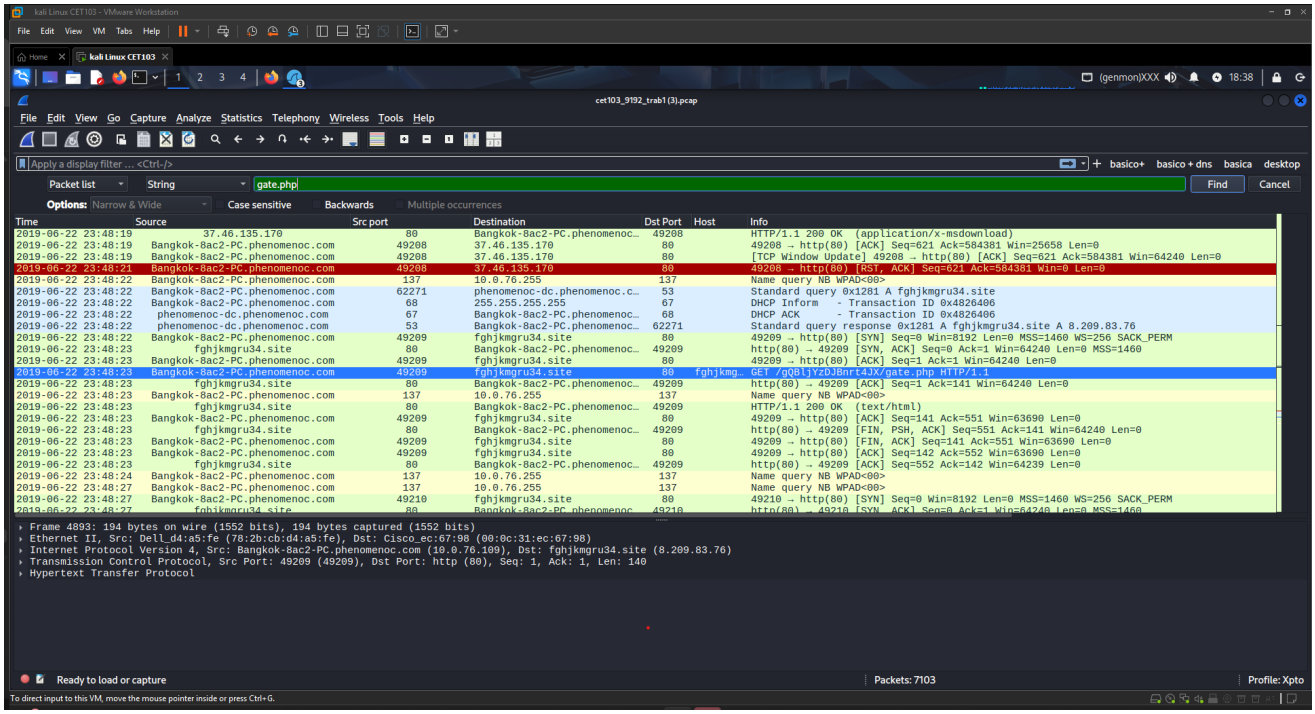
2.5 Review malware indicators and possible post-infection communications

The source investigates HTTP traffic containing `gate.php`, and reports the domain **fghjkmgru34.site** as an indicator of suspected post-infection traffic. The concluding section identifies **KPOT Stealer** as the malware family and mentions the address **8.209.83.76** in association with that domain. Treat these as source-reported indicators pending independent corroboration.

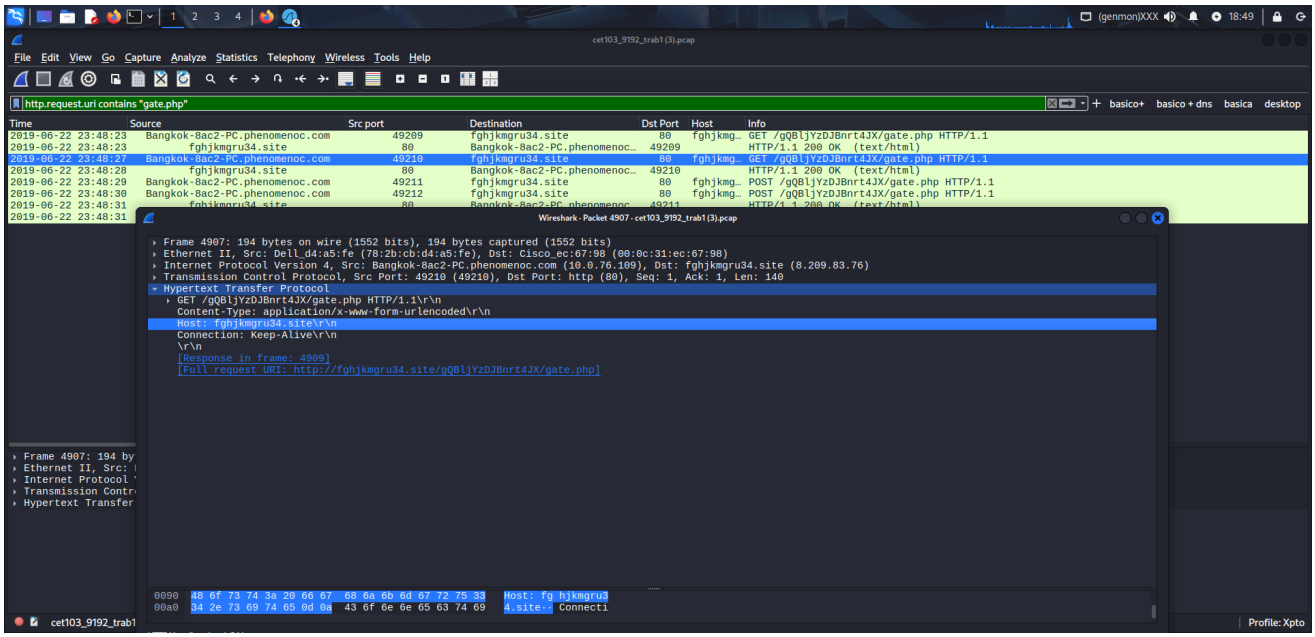
Useful display filter: `http.request.uri contains "gate.php"`. For a domain, inspect the HTTP Host field and, where available, DNS resolution in the capture. A matching string does not alone prove command-and-control; context and timing matter.



Evidence 08 — HTTP requests reviewed for gate.php



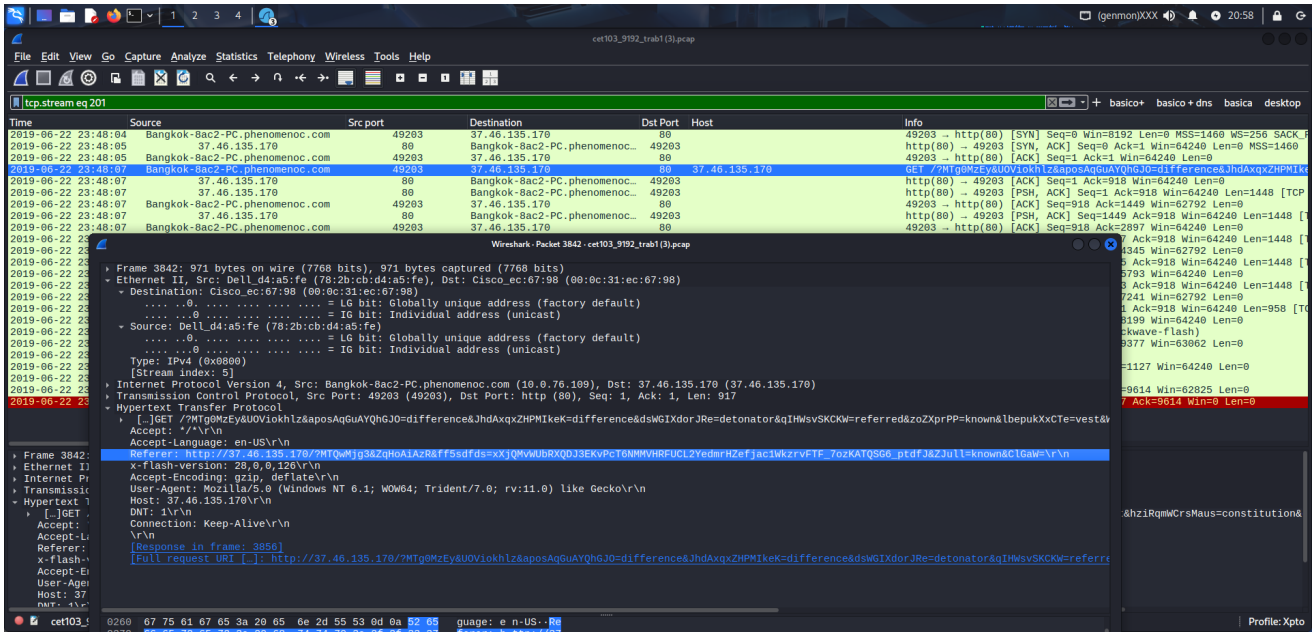
Evidence 09 — Suspicious host field and associated traffic



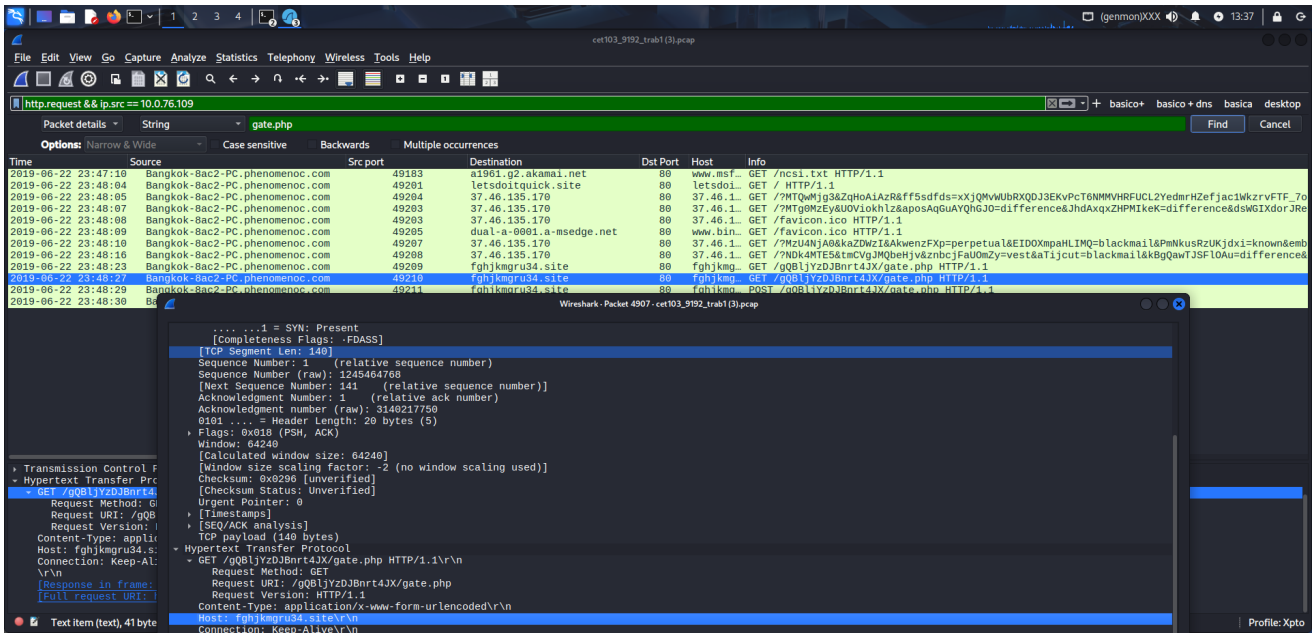
Evidence 10 — Follow TCP Stream or packet detail used for network correlation

2.6 Validate delivery infrastructure and domain relationships

The source associates the suspected delivery host with **37.46.135.170** and the subsequent suspicious domain with **fghjkmgru34.site**. The investigation distinguishes the server involved in payload delivery from a potentially different host observed after infection. This distinction is essential when writing accurate incident timelines.



Evidence 11 — Additional Wireshark investigation screenshot preserved from the source



Evidence 12 — Additional Wireshark investigation screenshot preserved from the source

3. Consolidated findings

Field	Source-reported value
Endpoint hostname	BANGKOK-8AC2-PC
Endpoint IPv4	10.0.76.109
Endpoint MAC	78:2b:cb:d4:a5:fe
Windows account	edris.haight
Suspected delivery IP	37.46.135.170
Suspected post-infection domain	fghjkmgru34.site
Associated IP in source conclusion	8.209.83.76
Reported malware family	KPOT Stealer / Rig EK delivery

Field	Source-reported value
SHA-256	Conflicting values; revalidation required

4. Evidence limitations and verification checklist

The available material is a report with screenshots rather than the underlying PCAP and extracted binary. Several screenshots are insufficient to verify every claim independently, and the reported SHA-256 values conflict. For a defensible forensic conclusion, re-open the original PCAP, locate the exact TCP stream and exported object, calculate SHA-256 from the preserved file, and correlate the network indicators with a contemporaneous threat-intelligence record.

Any extracted binary should be handled in an isolated analysis environment. Do not execute unknown samples on a workstation or upload confidential samples to public scanning services without authorization.

5. Conclusion

The documented analysis identifies a Windows endpoint, a named user account, suspicious HTTP delivery traffic, and an apparent post-infection domain. The case study demonstrates practical Wireshark use in incident triage and network forensics, including DHCP and Kerberos analysis, TCP stream reconstruction, HTTP object export and IoC documentation. Malware-family attribution and hash identification remain source-reported until the original evidence is independently re-examined.

References

Wireshark User’s Guide; NIST SP 800-61 (incident handling); ISO/IEC 27035 (incident management). References are methodological context and do not independently verify the case-specific findings.